

Peng Liu, Ph.D.

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📍 Boston



Research Interests

Medical Image Analysis, Deep Learning, Semi-supervised Learning

Education

2019 – 2024	Shanghai Jiao Tong University Ph.D. in Biomedical Engineering	Shanghai, China
	• Supervisor: Guoyan Zheng • Thesis: Semi-supervised Medical image Analysis Based on Consistency Regularization	
2015 – 2018	University of Chinese Academy of Science M.Sc. in Signal and Information Processing	Beijing, China
	• First place in the Postgraduate Entrance Examination • Thesis: Research on Object Detection in High-resolution Remote Sensing Image Based on Deep Learning	
2011 – 2015	China University of Geosciences B.Sc. in Geographic Information System	Beijing, China
	• GPA: 3.62/4.0 • Ranking: 8/60	

Employment History

Aug 2024 – Present	Boston College	Postdoctoral Researcher.
	• Advisor: Prof. Donglai Wei • Conduct research on Electron Microscopy (EM) image segmentation, focusing on mitochondrial segmentation, cell nuclei segmentation, and pan-organelle segmentation.	
Jul 2017 – Jul 2019	Meituan Group	Search and Recommendation Algorithms Engineer.
	• Participate in the optimization of tourism business ranking to improve the CTR(Click-Through-Rate) and CVR(Conversion Rate). • Related algorithm: Graph Embedding, DNN, Wide&Deep	

Research Publications(* indicates equal contribution)

Journal Articles

- 1 **P. Liu** and G. Zheng, "Handling imbalanced data: Uncertainty-guided virtual adversarial training with batch nuclear-norm optimization for semi-supervised medical image classification," *IEEE Journal of Biomedical and Health Informatics*, vol. 26, no. 7, pp. 2983–2994, 2022.

- 2 **P. Liu** and G. Zheng, "Cvcl: Context-aware voxel-wise contrastive learning for label-efficient multi-organ segmentation," *Computers in Biology and Medicine*, vol. 160, p. 106 995, 2023.
- 3 **P. Liu**, M. Zhang, X. Gao, B. Li, and G. Zheng, "Joint lymphoma lesion segmentation and prognosis prediction from baseline fdg-pet images via multitask convolutional neural networks," *IEEE Access*, vol. 10, pp. 81 612–81 623, 2022.
- 4 C. Chen* et al., "Diagnostic performance for severity grading of hip osteoarthritis and osteonecrosis of femoral head on radiographs: Deep learning model vs. board-certified orthopaedic surgeons," *Osteoarthritis Imaging*, vol. 3, no. 2, p. 100 092, 2023.
- 5 Z. Li, K. Chen, **P. Liu**, X. Chen, and G. Zheng, "Entropy and distance maps-guided segmentation of articular cartilage: Data from the osteoarthritis initiative," *International Journal of Computer Assisted Radiology and Surgery*, pp. 1–8, 2022.

Conference Proceedings

- 1 **P. Liu** and G. Zheng, "Semi-supervised learning regularized by adversarial perturbation and diversity maximization," in *Machine Learning in Medical Imaging: 12th International Workshop, MLMI 2021, Held in Conjunction with MICCAI 2021*, Springer, 2021, pp. 199–208.
- 2 **P. Liu** and G. Zheng, "Context-aware voxel-wise contrastive learning for label efficient multi-organ segmentation," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer, 2022, pp. 653–662.
- 3 Q. Zhou*, **P. Liu***, and G. Zheng, "Context-aware cutmix is all you need for universal organ and cancer segmentation," in *Fast, Low-Resource, and Accurate Organ and Pan-Cancer Segmentation in Abdomen CT*, Springer, 2023.
- 4 Q. Zhou, **P. Liu**, and G. Zheng, "Partially supervised multi-organ segmentation via affinity-aware consistency learning and cross site feature alignment," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer, 2023, pp. 671–680.

Under Review

- 1 **P. Liu** et al., "Mitoem 2.0: A benchmark for challenging 3d mitochondria instance segmentation from em images," Under Review at Scientific Data, 2025.
- 2 **P. Liu** and G. Zheng, "Cr-gloco: Cross-resolution learning via global-local context consistency for semi-supervised 3d medical segmentation," Minor Revision at Computerized Medical Imaging and Graphics, 2025.

In Preparation

- 1 **P. Liu** and D. Wei, *Skelaffinity-seg: Skeleton-guided geometry and global affinity learning for complex 3d tubular instance segmentation*, To be submitted to IEEE Transactions on Medical Imaging, 2026.
- 2 **P. Liu** and D. Wei, *Mitoverse: A foundation model for mitochondrial morphology, topology, and instance segmentation*, To be submitted to IEEE Transactions on Medical Imaging, 2026.
- 3 **P. Liu** and D. Wei, *Mixture-of-experts for multi-resolution pan-organelle segmentation in 3d volume electron microscopy*, To be submitted to IEEE Transactions on Medical Imaging, 2026.

Competitions

Scholarships

Teaching Assistant

Peer Reviewer

Skills

Languages	Good reading, writing and speaking competencies for English. (CET-6, TOEFL)
Coding	Linux (7 yrs), C++ (10 yrs), Python (7 yrs), PyTorch (6 yrs)